

# Length-Controlled News Summarization Using Mistral-7B with LoRA Fine-Tuning

Fine-tuning Small Language Models for News Summarization Task

## 1. Team Details

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## 2. Introduction

Abstractive text summarization remains a challenging task for large language models (LLMs), particularly when strict output constraints such as summary length and factual faithfulness must be enforced simultaneously. While recent instruction-tuned models demonstrate strong zero-shot summarization performance, their deployment is often infeasible in low-resource environments due to high computational and memory requirements.

This work investigates the feasibility and effectiveness of **instruction-tuned small language models (SLMs)** for **length-controlled abstractive news summarization** under **severely constrained computational settings**. Specifically, we fine-tune **Mistral-7B-Instruct** using **Low-Rank Adaptation (LoRA)** on a reduced subset of the CNN/DailyMail dataset. The primary goal is to analyze how **training time (epochs)** and **limited data volume** affect:

- Instruction adherence (length control),
- Summary fluency and coherence,
- Hallucination behavior, and
- Factual faithfulness to the source article.

Due to GPU memory and execution-time constraints, training was deliberately restricted to a small dataset subset and a small number of epochs. This controlled setup enables a focused comparison between **early-stage undertrained models** and **later-stage partially adapted models**, highlighting how hallucination behavior evolves during fine-tuning.

### 3. Experimental Setup

- **Platform:** Google Colab (Free Tier with GPU support)
- **Fine-tuning Framework:** Unsloth
- **Fine-tuning Method:** Parameter-Efficient Fine-Tuning (PEFT) - QLoRA
- **Base Infrastructure:** HuggingFace Hub for model hosting and sharing

This setup enabled rapid iteration and resource optimization, showcasing the feasibility of advanced NLP tasks on accessible hardware without significant computational overhead.

### 4. Model and Training Configuration

#### 4.1 Base Model

- **Model:** mistralai/Mistral-7B-Instruct-v0.2
- **Architecture:** Decoder-only transformer
- **Pretraining:** Large-scale web and instruction-following corpora
- **Quantization:** 4-bit NormalFloat (NF4)
- **Computation Precision:** FP16
- **Fine-Tuning Method:** Parameter-Efficient Fine-Tuning (LoRA)

The choice of Mistral-7B-Instruct reflects a balance between strong instruction-following capabilities and manageable computational cost. **NF4 quantization** significantly reduced memory footprint, enabling fine-tuning on consumer-grade GPUs. However, quantization can amplify instability during training, particularly in early epochs, potentially exacerbating hallucination in undertrained regimes.

#### 4.2 Fine-Tuning Strategy

- **Training Paradigm:** Supervised Fine-Tuning (SFT)
- **Dataset:** CNN/DailyMail (v3.0.0)
- **Training Samples:** ~1,000 articles
- **Validation Samples:** ~200 articles
- **Epochs:** 3
- **Optimizer:** AdamW (8-bit)
- **Batching:** Gradient accumulation to simulate larger effective batch sizes

#### LoRA Configuration

LoRA adapters were injected into the model’s attention projection layers, allowing task-specific adaptation while freezing the majority of pretrained parameters. This significantly reduced trainable parameters and stabilized training under limited compute, but also constrained the model’s ability to deeply relearn content selection and factual grounding.

## Instruction Design

Each training instance followed an explicit instruction template requiring summaries of **exactly 2, 3, or 5 sentences**. This explicit conditioning enabled a direct assessment of the model’s ability to disentangle **structural obedience (length control)** from **semantic fidelity (faithfulness)**.

## 5. Inference and Evaluation Setup

### 5.1 Inference Configuration

- **Decoding:** Autoregressive generation
- **Sampling:** Enabled (do\_sample=True)
- **Temperature:** 0.7
- **Top-p:** 0.9
- **Repetition Penalty:** 1.2
- **Maximum Tokens:** 90 (for medium-length summaries)

This decoding configuration was selected to balance readability and lexical diversity. However, stochastic decoding is known to **increase hallucination risk**, particularly in undertrained or weakly grounded models. As such, hallucination observed in early epochs can be partially attributed to the interaction between sampling and insufficient fine-tuning.

### 5.2 Evaluation Metrics

To capture both surface-level similarity and deeper semantic alignment, we employed a combination of lexical and embedding-based metrics:

#### Lexical Overlap Metrics

- **ROUGE-1**
- **ROUGE-2**
- **ROUGE-Lsum**

These metrics measure n-gram overlap and structural similarity to reference summaries, serving as proxies for content coverage and factual alignment.

#### Semantic Similarity Metric

- **BERTScore (RoBERTa-large)**

BERTScore evaluates contextual semantic similarity using pretrained embeddings. While effective at measuring fluency and topical relevance, it is **insensitive to factual hallucinations**, making it insufficient as a standalone faithfulness metric.

This dual-metric setup highlights cases where fluent but factually incorrect summaries achieve deceptively high semantic scores.

## 6. Experimental Results and Observations

### 6.1 Epoch-Wise Behavioral Analysis

#### Epoch 1: Undertrained Model

##### Qualitative Observations

6. The model largely respected sentence-count constraints.
7. Summaries frequently introduced **unsupported entities and events**, such as:
  - a. Mentions of Kanye West’s shoe sales,
  - b. Fabricated prices or unrelated commercial details,
  - c. Events absent from the source article.
8. The summary often drifted away from the central news narrative.

##### Hallucination Characteristics

- Hallucinations were **topically plausible but factually incorrect**.
- Generated content reflected **pretrained world knowledge** rather than article-conditioned reasoning.
- Indicates weak alignment between input text and output generation.

##### Interpretation

The large discrepancy between ROUGE and BERTScore indicates that the model produced semantically fluent summaries that were **poorly grounded in the source text**. This confirms that high semantic similarity does not guarantee factual correctness.

### 6.2 Epoch 3: Better-Trained Model

##### Qualitative Improvements

- Significant reduction in fabricated entities and events.
- Improved alignment with article-specific facts.
- More consistent inclusion of key information.
- Enhanced sentence-to-sentence coherence.

### Behavioral Shift

The model transitioned from **prior-driven generation** to **input-conditioned summarization**, demonstrating improved grounding despite the limited dataset size.

## 7. Effect of Data Size and Training Time

### 7.1 Impact of Limited Data

With only ~1,000 training examples:

- The model effectively learned:
  - Instruction format,
  - Sentence-count control,
  - General summarization structure.
- However, it struggled with:
  - Accurate content selection,
  - Faithful abstraction of article details.

This resulted in summaries that were structurally correct but **content-incomplete or partially distorted**.

### 7.2 Impact of Training Time (Epochs)

| Training Setting           | Observed Behavior                         |
|----------------------------|-------------------------------------------|
| Fewer epochs               | High hallucination, weak grounding        |
| 3 epochs                   | Reduced hallucination, improved alignment |
| More epochs (hypothesized) | Higher ROUGE, stronger faithfulness       |

Training duration had a **stronger influence on hallucination reduction** than dataset size alone, emphasizing the importance of sufficient optimization even under data scarcity.

## 8. Length Control vs. Faithfulness Trade-Off

A central finding is that **length control is learned significantly faster than factual faithfulness**.

- The model consistently obeyed:
  - “2 sentences”
  - “3 sentences”
- However:
  - Sentence correctness,

- Content accuracy,
- Evidence grounding lagged behind.

This demonstrates that:

- Instruction obedience is a **surface-level behavior**,
- Faithfulness requires deeper representation updates and more exposure to aligned data.

## 9. Limitations

### Compute Constraints

- Small dataset subset
- Limited number of epochs

## 10. Conclusion

This study demonstrates that instruction-tuned small language models can perform length-controlled news summarization effectively under limited computational resources. However, when training data and epochs are insufficient, models tend to hallucinate by injecting pretrained knowledge rather than grounding summaries in the source text.

## 11. References

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|----------------------------------------------|------------------|------------|----------|----------|---------|-------|
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